Question Paper - Report

18-Mar-2025 07:04:15
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Question Paper

Back

MANIPAL ACADEMY OF HIGHER EDUCATION

ALGEBRAIC STRUCTURES [MAT 2235]

Marks: 30 **Duration: 90 mins.**

MCQs

Answer all the questions. Section Duration: 20 mins

1) With usual notation, in the dihedral group of order 12, $(sr^9)(sr^6)$ is _____. (1)

r^8 r^{10} r^7 r^9

2) Let $f:(\mathbb{R},+)\to(\mathbb{C}\setminus\{0\},\times)$ be the homomorphism defined by $f(x)=\cos x+i\sin x$. Then $\ker f$ is ____. (1)

$\{n\pi : n\in\mathbb{Z}\}$ $\{2n\pi : n\in\mathbb{Z}\}$ $\left\{(2n+1)\frac{\pi}{2} : n\in\mathbb{Z}\right\}$ $\{0\}$

3) Write the permutation (12345) as a product of transpositions. (1)

$(15)(12)(14)(13)$ $(15)(14)(13)(12)$ $(12)(34)(51)$ $(12)(13)(14)(15)$

4) Let $\mathbb{R}$ be the set of real numbers operation $*$ defined by $a*b=a+b+1$. Then in the group $(\mathbb{R},*)$, the identity is _____. (1)

1 -1 0 -2

5) Let G be a group. Then by solving simultaneously the equations $x^2=b$ and $x^5=e$, the value of x is _____. (1)

Inverse of b Inverse of b^2 Inverse of a Inverse of a^2

Descriptive

Answer all the questions.

6) Prove that any infintie cycle group is isomorphic to $\mathbb{Z}$. (2)

7) Let $G=\{e,a,b,b^2,ab,ab^2\}$ with $a^2=e$, $b^3=e$, $ba=ab^2$. Draw a Cayley diagram representation of G. (2)

8)	Compute the distinct left cosets of the group $(\mathbb{Z}_{12}, +)$ with respect to the subgroup $H = \{0,6\}$.	(2)
9)	If $a \in Hb$, then prove that $Ha = Hb$. Hence show that the family of all cosets Ha, as a ranges over G, is a partition of G.	(2)
10)	Let G be a group acting on a set S, and $s \in S$ be fixed. Prove that the stabilizer of s in G (that is, $Gs = \{g \in G : g \cdot s = s\}$) is a subgroup of G.	(3)
11)	Let $U(n)$ be the multiplicative group of integers less than n and relatively prime to n. Then, with usual notation, compute the left regular representation $\overline{U(18)}$ for $U(18)$.	(3)
12)	Prove that the center of a group is a normal subgroup. Compute the center of $D_8 = \langle r, s : r^4 = 1 = s^2, rs = sr^{-1} \rangle$ the dihedral group of order 8.	(3)
13)	Prove that every group is isomorphic to a group of permutations.	(4)
14)	Prove that a subgroup of a cyclic group is cyclic. Is every cyclic group Abelian?	(4)

^ Top